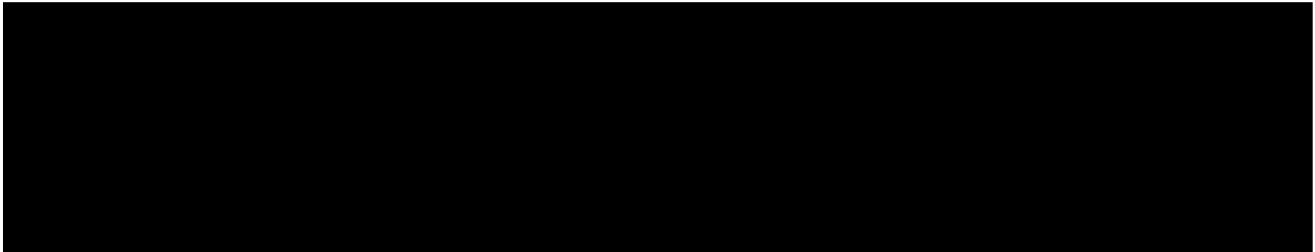


Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.



Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

This program will conduct spectroscopy on a supernova using the “B&C” Spectrograph on the Bok 90” telescope at Kitt Peak, Arizona. Due to the transient nature of supernovae, needing to be observed within approximately thirty days of peak brightness, the exact scientific goal of the project will not be known until the observations are conducted in Mid-October 2026. However, we can anticipate that the spectroscopy might allow us to categorize the supernova if it is caught early enough, which would be a significant scientific result. This would be done by fitting the spectrum with the SNID-Sage software to identify defining features of supernova types such as the presence of Hydrogen and/or Helium lines. Regardless of the supernova type, fitting the spectrum with the SNID-Sage software will generate interesting scientific results like allowing us to estimate the radial velocity of the target from the Doppler shift of spectral lines and the relative abundances of various elements within the target. This proposal offers a sample outline and representative targets, but the ultimate target and science goal will be determined by availability at the time of observing.

Telescope Requested (0.5 points): BOK 90”

Instrument (0.5 points): “B&C” Spectrograph

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

At the time of our observation (Oct. 17-19), the sun will have an RA of around 14 hours, while the moon's RA will be between 18 and 20. This is not ideal for our observations as this means the moon will be reaching its peak right around sunset, but via the Catalina Sky Survey we can anticipate the brightness on these dates, from 38% to 58% illuminated as the weekend progresses. Thus, it would be more ideal to conduct our observations on October 17th if possible, as this is when the moon will be least bright and thus will have the least interference with our observations. The moon will set between 11:00PM and 1:00AM on all of these dates, so it may be advantageous to observe a target with an RA around 22-24 or 0-3 hours as these will be crossing the meridian while both the sun and moon are below the horizon.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

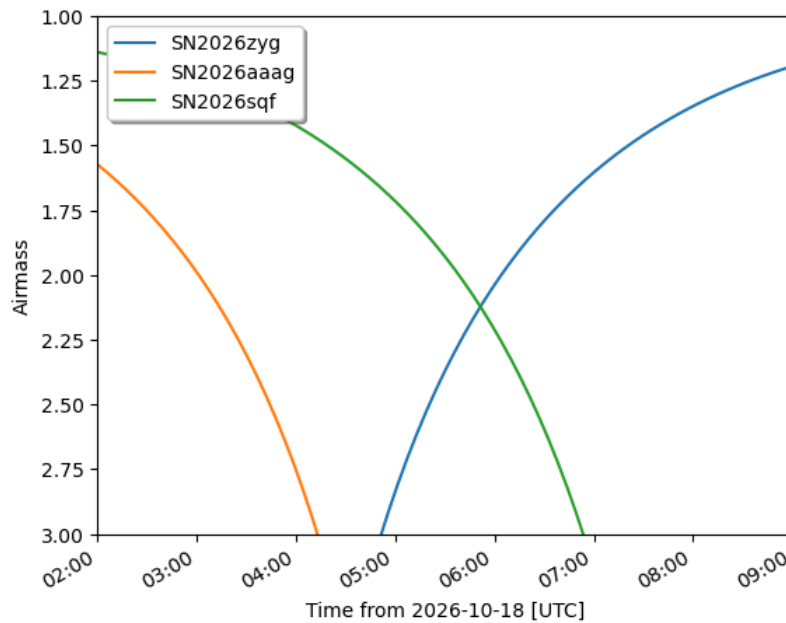
[Database of Recently Discovered SN](#)

Constellations highly visible in Oct: Cygnus, Pegasus, Cassiopeia, Cephus, Andromeda, Ursa Major/Minor

Constellations only visible at beginning/end of night in Oct: Aquarius, Capricornus, Cetus, Cygnus, Aquila, Lyra, Taurus, Auriga, Orion, Lynx

Object Name	Right Ascension	Declination	V magnitude at peak	Peak Date	Object Type	Visibility
SN 2026sqf	140.673888	53.37	12.9	7/28	SN Ia	ursa major
SN 2026aaag	102.05	51.23555556	18.08	8/29	SN type SN II	Lynx

SN 2026zyg	308.8761	51.5900	18.049	8/28	SN II	Cygnus
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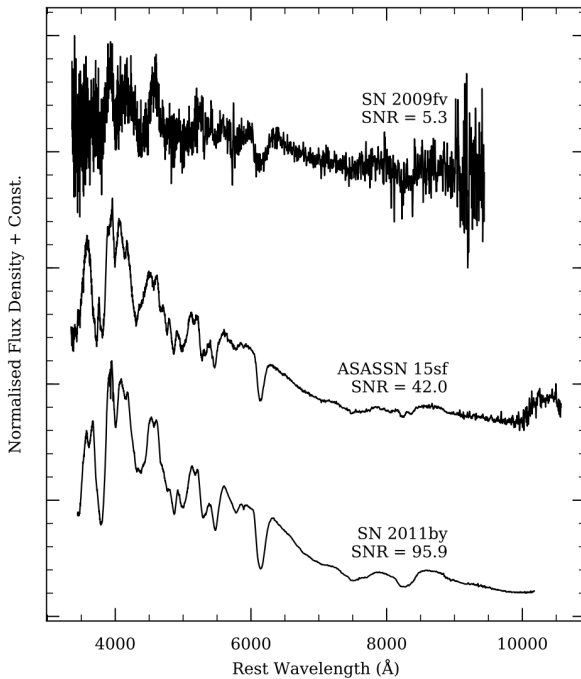


The plot shows the airmass for the targets outlined in the table for the night of October 17th from 7:00PM to 2:00AM Tucson time. The line in blue would be a better choice for a target in our case because it reaches peak later in the night when the moon will be set, and it gets very high in the sky which improves viewing with less airmass.

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

In order to discern distinct spectral features like Hydrogen and Helium within the supernova, we will want a minimum signal to noise ratio of five, ideally higher. This is based on a paper from Stahl et. al (2020) which observed Type Ia supernovae at varying signal to noise ratios and illustrated the visibility of spectral features for each. They used a 3 meter telescope, and in the paper they cite an SNR of 8.3 with an exposure time of 1200 seconds for one of their observations, and some much higher SNRs for the same exposure time. With a 2.3 meter telescope and probably a similarly bright source, we would likely need around 1200 seconds of exposure time to get an SNR of 5.



How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

Taking a spectrum of the supernova will allow us to measure chemical absorption/emission lines, which are used in the categorization of the supernova based on the elements present from their features. For example, if there are no Hydrogen features in the spectrum, we will know that it is a Type I supernova, and the ionized Silicon (Si II) feature is an indicator of a type Ia supernova specifically. Using the SNID-Sage program will fit the spectrum and easily identify these features, while computations like optical depth can give us an idea of the abundances of certain elements within the supernova. We can estimate the radial velocity from the Doppler shift of certain known elemental features.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If we are unable to gather our own spectroscopic data, we may be able to collect similar observations from projects that have been conducted by science teams at the University of Arizona recently. For example, Professor Nathan Smith does regular supernova spectroscopy using the same instrument, including data as recent as August of this year, so we could ask him to use some of his observations if we cannot get our

own. That research group took spectra of a type Ic broad-line supernova earlier this year, which would be a unique and interesting science target for an alternative project.